

Transport-projected exact diagonalization of quantum spin ice: benchmarks and convergence criteria

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Exact diagonalization of quantum spin ice on a periodic cluster can generate noncontractible four-spin tunneling at second order, parametrically before the physical six-spin ring exchange. We formulate a nonperturbative remedy: exactly diagonalize a transported-dipole source family, canonically pull its isolated ice-descended band to a common basis, and project its zero Fourier component. At $J_{\pm}/J_{zz} = 0.046$, the character sequence converges from $M = 3$ to $M = 4$ within 0.50% in centered operator norm, with minimum ice overlap 0.762. Projection moves the cubic-16 heat-capacity peak from $0.0255J_{zz}$ to $0.00146J_{zz}$, compared with the quantum Monte Carlo value $0.00110J_{zz}$, while retaining the exact microscopic complement. The low-temperature heat-capacity error is reduced by 72.9% but remains just outside our preregistered tolerance, and entropy also fails. Thus the method is nonperturbatively converged on cubic-16, but the cluster is not yet a quantitative surrogate for bulk quantum spin ice.

Motivation.—Quantum spin ice (QSI) realizes a compact $U(1)$ gauge theory in a pyrochlore magnet: the two-in-two-out constraint is Gauss’s law, defects are electric charges, and collective hexagon tunneling generates the emergent photon [1–3]. This low scale is experimentally consequential. In particular, the two heat-capacity features reported in $\text{Ce}_2\text{Hf}_2\text{O}_7$ motivate calculations that retain both the spin-ice crossover and quantum dynamics [4]. Numerically, the problem is divided by sign. Quantum Monte Carlo (QMC) reaches the zero-flux XXZ model and resolves its thermodynamics and dynamics [5, 6], whereas the opposite sign relevant to π -flux candidates has a sign problem. Exact diagonalization (ED) has no sign problem, but its periodic clusters alter the shortest allowed gauge processes.

We address that finite-size problem without replacing the microscopic model by an effective Hamiltonian. Perturbation theory is used only to identify the offending path and test its topology; production results come from exact microscopic eigenspaces. The controlled numerical limit is the transport-character resolution $M \rightarrow \infty$. On the sign-free side, cleaning must also improve convergence toward QMC. Exact removal of a known artifact is not by itself proof of convergence to the bulk.

The finite torus changes the leading process.—We study

$$H = J_{zz} \sum_{\langle ij \rangle} S_i^z S_j^z - J_{\pm} \sum_{\langle ij \rangle} (S_i^+ S_j^- + S_i^- S_j^+), \quad J_{zz} > 0. \quad (1)$$

The Ising ground states form the ice manifold P . In the infinite pyrochlore lattice the shortest contractible ice-preserving cycle is a hexagon. Its third-order matrix element has scale $g_6 = 12|J_{\pm}|^3/J_{zz}^2$ [1]. On a torus, however, a path may close through a periodic image. Cubic-16 contains 36 winding four-cycles in two winding classes [Figs. 1(a) and 1(b)] and 16 contractible hexagons [Fig. 1(c)]. The four-spin process enters at second order

with $g_4 = 4J_{\pm}^2/J_{zz}$, so $g_4/g_6 = J_{zz}/(3|J_{\pm}|)$ diverges in the nominally controlled limit. FCC-32 reduces but does not eliminate the problem: it has 24 winding four-cycles, 32 contractible hexagons, and 96 wrapping hexagons.

Nonperturbative transport projection.—Let P project onto the 90-state cubic-16 ice manifold. Attach a microscopic source to every spin exchange so that a completed path carries $e^{-i\theta \cdot \mathbf{q}}$, where the primitive integer transported dipole is $\mathbf{q} = 2\delta$. For every θ , we diagonalize the microscopic $H(\theta)$ in the $S^z = 0$ sector. If P_{θ} is the isolated 90-state spectral projector, the polar map

$$W_{\theta} = P_{\theta} P (P P_{\theta} P)^{-1/2} \quad (2)$$

canonically identifies it with P . The exact pulled-back band is $h(\theta) = W_{\theta}^{\dagger} H(\theta) W_{\theta}$. We define

$$h_M = \frac{1}{M^3} \sum_{\mathbf{m} \in \mathbb{Z}_M^3} h(2\pi \mathbf{m}/M). \quad (3)$$

Equation (3) retains $\mathbf{q} = 0 \pmod{M}$; nonperturbatively, $M \rightarrow \infty$ is required. A singular $P P_{\theta} P$ is an explicit failure of the method. The canonical Schrieffer–Wolff series [7] recovers this construction order by order and supplies the loop topology test, but it is not used to construct the reported spectrum. Thus the operation is neither a fitted counterterm nor an average of partition functions. For thermal observables we replace only the bare band by the projected band,

$$Z_{\text{clean}} = Z_{\text{full}} - Z_{\text{bare band}} + Z_{\text{clean band}}, \quad (4)$$

and apply the same identity to the first two energy moments. This is the spectrum of one Hermitian, temperature-independent operator and leaves the microscopic complement intact. Derivations and numerical details are given in the Supplemental Material [8].

Internal tests.—Direct row enumeration through third order gives a binary topology test. On both clusters, Eq. (3) removes 100% of winding four-loop weight

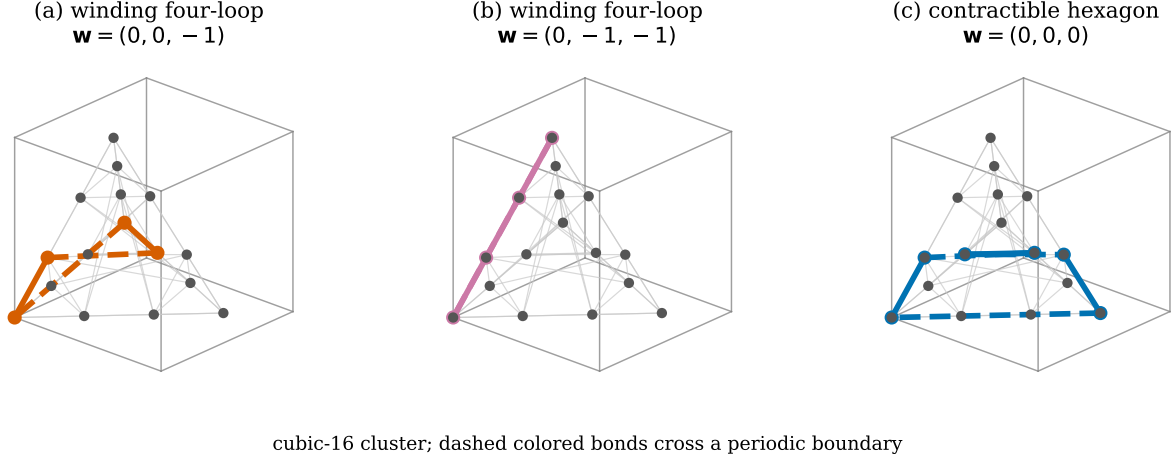
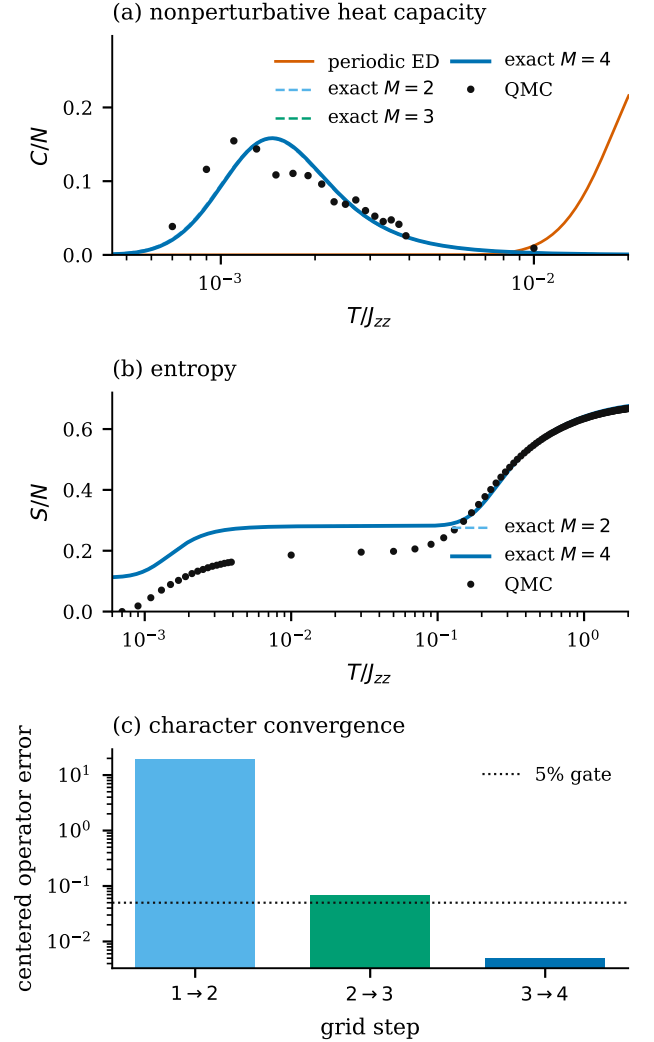


FIG. 1. Representative gauge processes hosted on the periodic cubic-16 pyrochlore cluster. (a),(b) The two classes of ice-preserving four-spin loops close only through periodic identification. (c) The physical six-spin hexagon closes in the covering lattice. Dashed colored bonds cross the chosen cell boundary; the winding vector $\mathbf{w}$ is basis dependent, while its vanishing is not.

and wrapping-hexagon weight and retains 100% of contractible-hexagon weight. At $J_{\pm}/J_{zz} = 0.046$, cubic-16 moves from the unprojected $T_p/J_{zz} = 0.02554$ to 0.001252. FCC-32 gives 0.008770 and 0.000608 respectively within the same third-order diagnostic. These FCC-32 values are not production thermodynamics.



The exact cubic-16 calculation uses one source convention at every M . The zero-source band has minimum ice overlap 0.762 and fixed- S^z gap $0.627J_{zz}$. All $M = 3, 4$ source points have larger overlaps and gaps, with maximum Ritz residual 3.7×10^{-12} . The centered operator changes by 6.78% from $M = 2$ to 3, but only 0.498% from $M = 3$ to 4, passing the 5% character gate [Fig. 2(c)].

FIG. 2. Nonperturbative validation at $J_{\pm}/J_{zz} = 0.046$ against Huang *et al.* [6]. (a) Exact character-projected heat capacity after retaining the microscopic complement. (b) Entropy remains discrepant. (c) The $M = 3 \rightarrow 4$ centered-operator change passes the 5% gate.

External QMC gate.—Huang *et al.* simulate precisely Eq. (1) at $J_{\pm}/J_{zz} = 0.046$ using continuous-time worm QMC and report thermodynamics together with QMC-SAC dynamics on 8^3 primitive cells [6]. A vector extraction of their curves gives low- and high-temperature peaks at $T/J_{zz} = 0.001100$ and 0.2105 . The converged $M = 4$ result places the low peak at $0.001463J_{zz}$ and reduces the low-temperature heat-capacity RMSE from 0.0859 to 0.02327 , a 72.9% improvement. It nevertheless misses the preregistered 0.02 tolerance. Entropy also fails: its RMSE is 0.04509 , again above 0.02 [Fig. 2(b)]. The same paper provides the next dynamics gate, S^{zz} and S^{+-} at $T/J_{zz} = 0.001, 0.04$, and 0.1 ; matching its momentum points, normalization, and SAC resolution remains to be done.

Outlook.—Transport projection solves a sharply defined problem: it removes nonzero-winding processes from a canonically identified gauge band. The exact $M = 3, 4$ agreement demonstrates that this correction is not merely a low-order perturbative construction. It also shows why convergence of the method must be separated from convergence of the cluster: cubic-16 remains outside the QMC heat and entropy tolerances. A material application therefore requires a full-Hamiltonian FCC-32 calculation, cluster convergence, and matched S^{zz} and S^{+-} dynamics before entering the π -flux regime.

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 - [8] (2026), see Supplemental Material for the canonical block construction, transported-dipole projection, exact-band limit, thermodynamic replacement identities, geometry, and benchmark provenance.